



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design) Practice 3: Projective Geometry and Camera Model

Spring 2026

Instructor: Hao Li

Outline

- Wonder of Projective Geometry
- Pinhole Camera Model
- Use of Homography

《墨子·经说下》

景，光之人煦若射，
下者之人也高，高者之人也下。
足敝下光，故成景于上；
首敝上光，故成景于下。
在远近有端，与于光，
故景庠（障）内也。

景到，在乎有端，与景长，
说在端

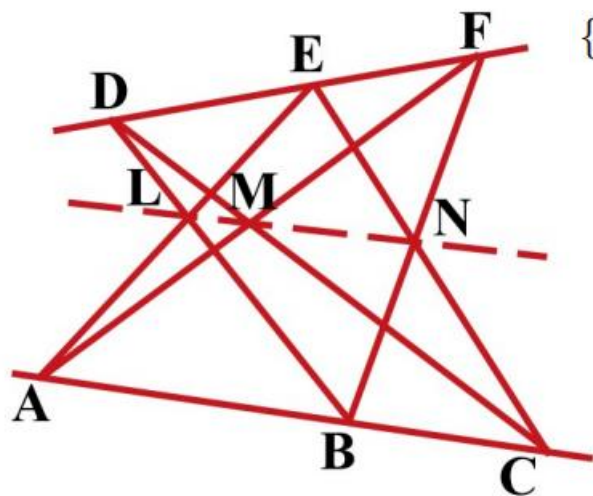


THREE-DIMENSIONAL COMPUTER VISION



Perspective painting technique

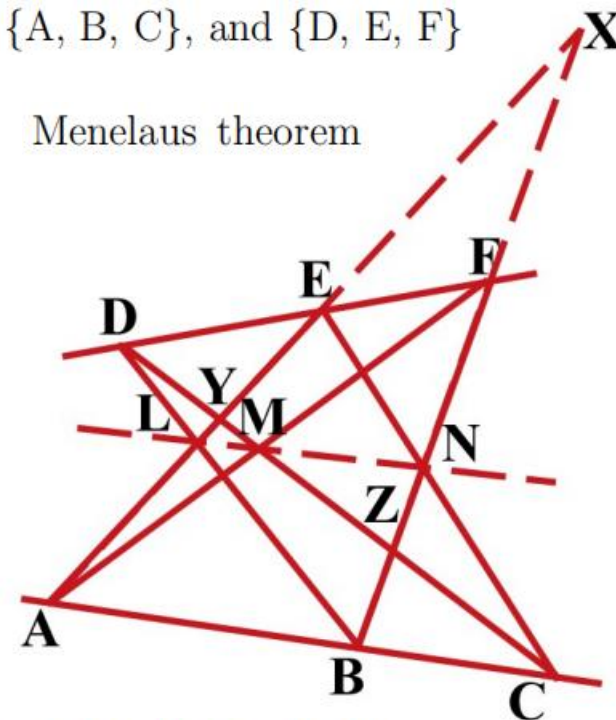
Pappus Theorem



$\{A, M, F\}$, $\{B, L, D\}$, $\{C, N, E\}$, $\{A, B, C\}$, and $\{D, E, F\}$

$$\begin{aligned} \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} &= 1, \\ \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} &= 1, \\ \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} &= 1, \\ \frac{|YC|}{|CZ|} \cdot \frac{|ZB|}{|BX|} \cdot \frac{|XA|}{|AY|} &= 1, \\ \frac{|XE|}{|EY|} \cdot \frac{|YD|}{|DZ|} \cdot \frac{|ZF|}{|FX|} &= 1. \end{aligned}$$

Menelaus theorem



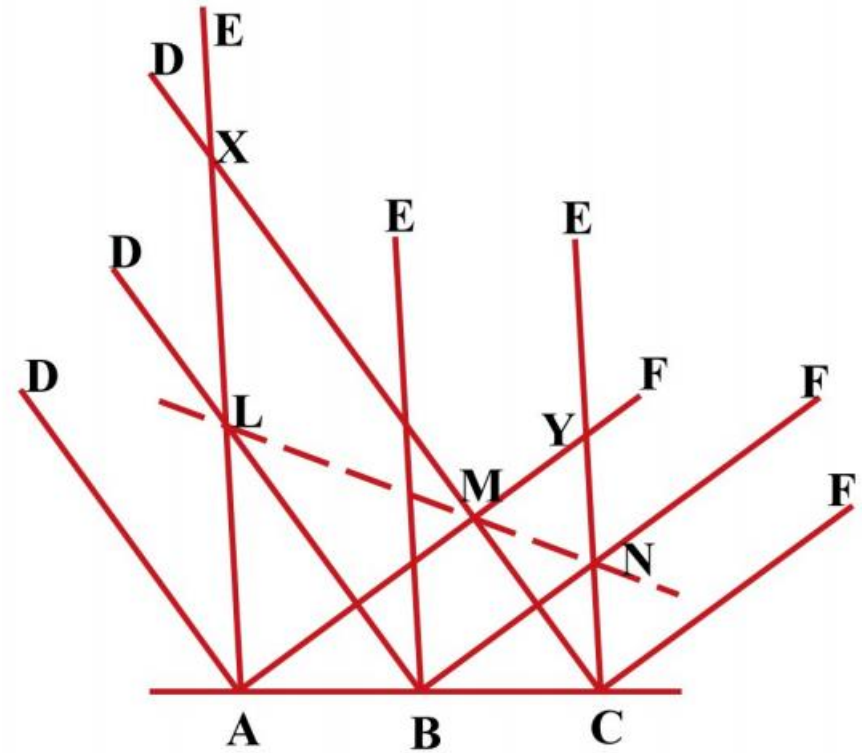
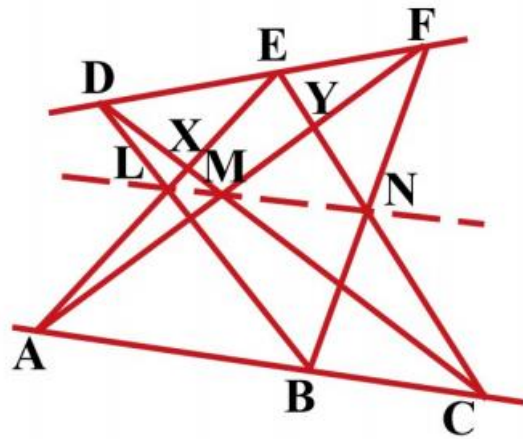
$$\begin{aligned} \prod \text{left sides} &= \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} \cdot \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} \\ &\quad \cdot \frac{|YC|}{|CZ|} \cdot \frac{|ZB|}{|BX|} \cdot \frac{|XA|}{|AY|} \cdot \frac{|XE|}{|EY|} \cdot \frac{|YD|}{|DZ|} \cdot \frac{|ZF|}{|FX|} \\ &= \frac{|ZM|}{|MY|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XN|}{|NZ|} = 1. \end{aligned}$$

Invariance In Projective Geometry

- *Invariance 1*: A point after perspective mapping (or perspective transform) is still a point.
- *Invariance 2*: A line after perspective mapping is still a line.
- *Invariance 3*: Coinciding lines after perspective mapping still coincide.
- *Invariance 4*: Colinear points after perspective mapping are still colinear.



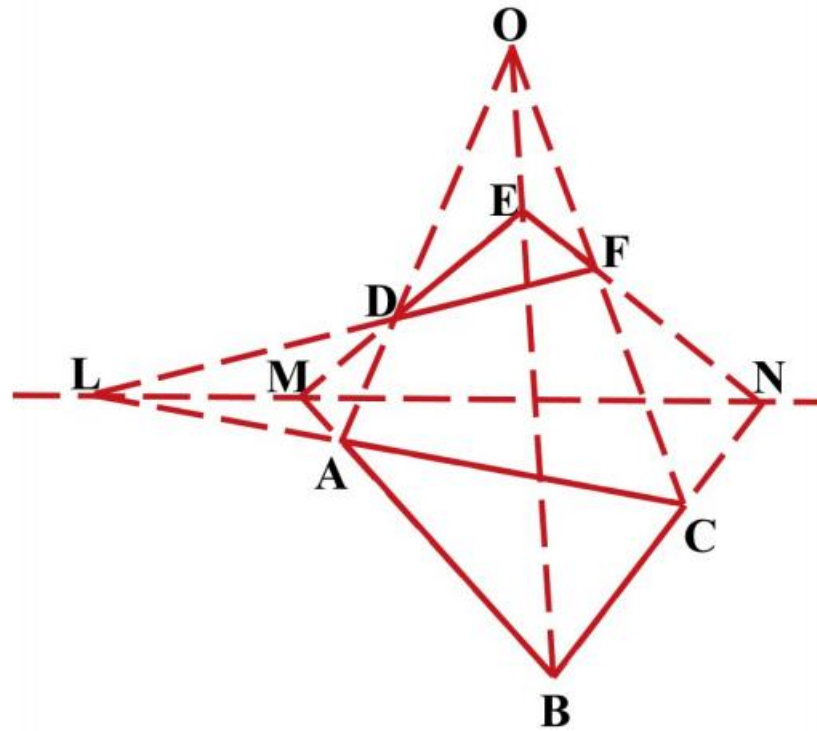
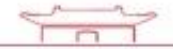
Pappus Theorem - PG Perspective



$$\frac{|XL|}{|LA|} = \frac{|CB|}{|BA|} = \frac{|CN|}{|NY|}$$

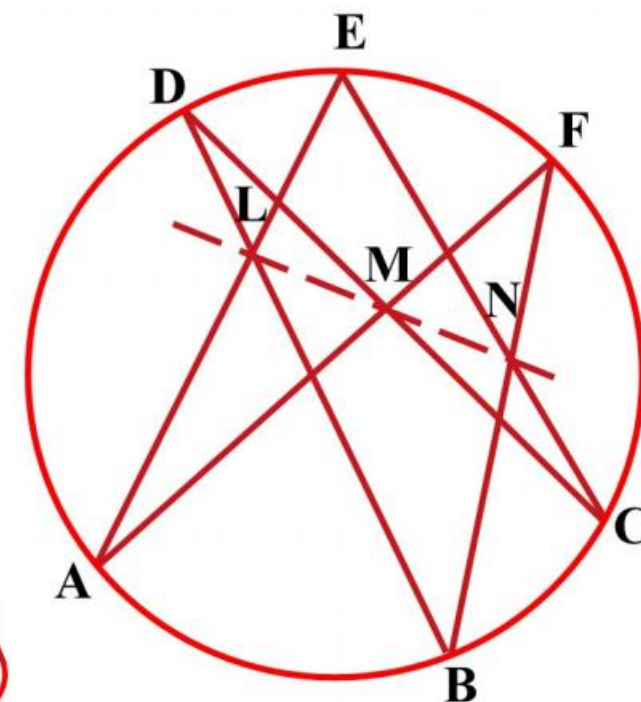
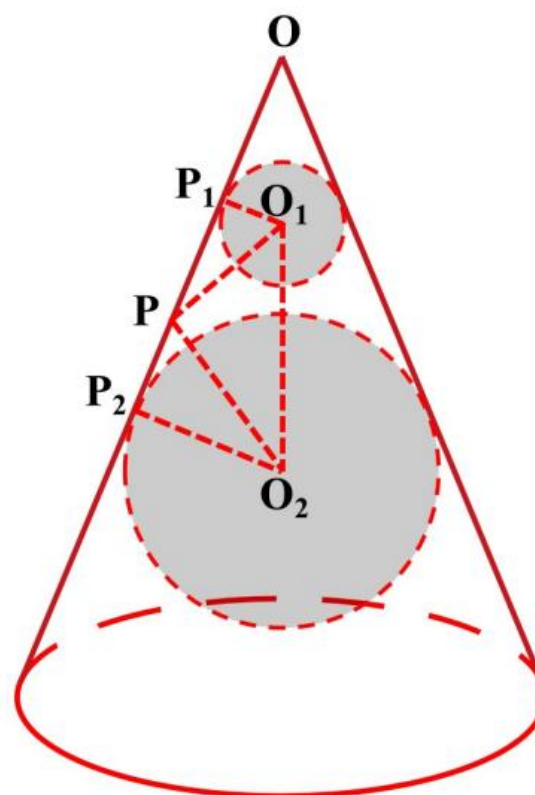
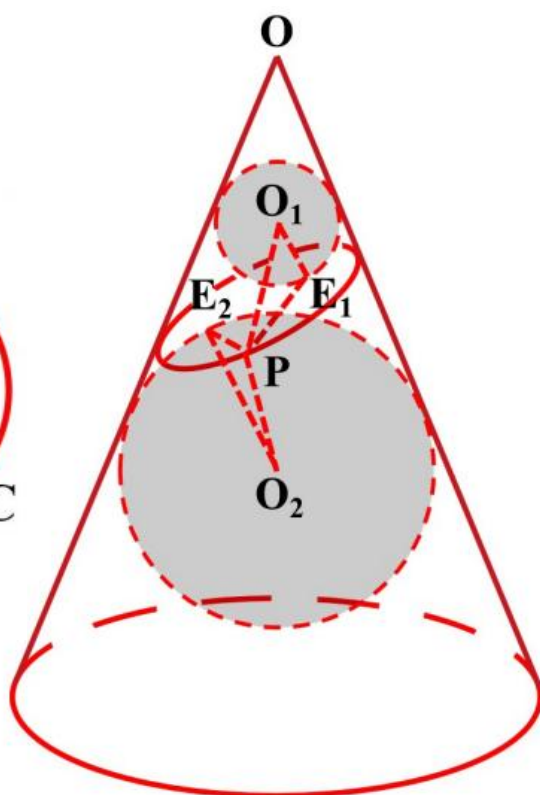
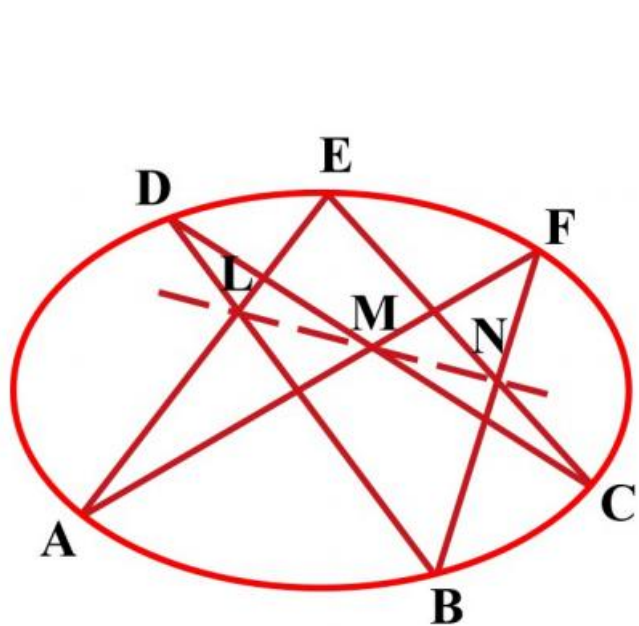


Desargue Theorem - PG Perspective



of a tetrahedron intersected by two planes in the three-dimensional space. The tetrahedron is formed by the four triangles O - A - B , O - B - C , O - C - A , and A - B - C . The two planes are spanned by the triangle A - B - C and the triangle D - E - F respectively.

Pascal Theorem - PG Perspective

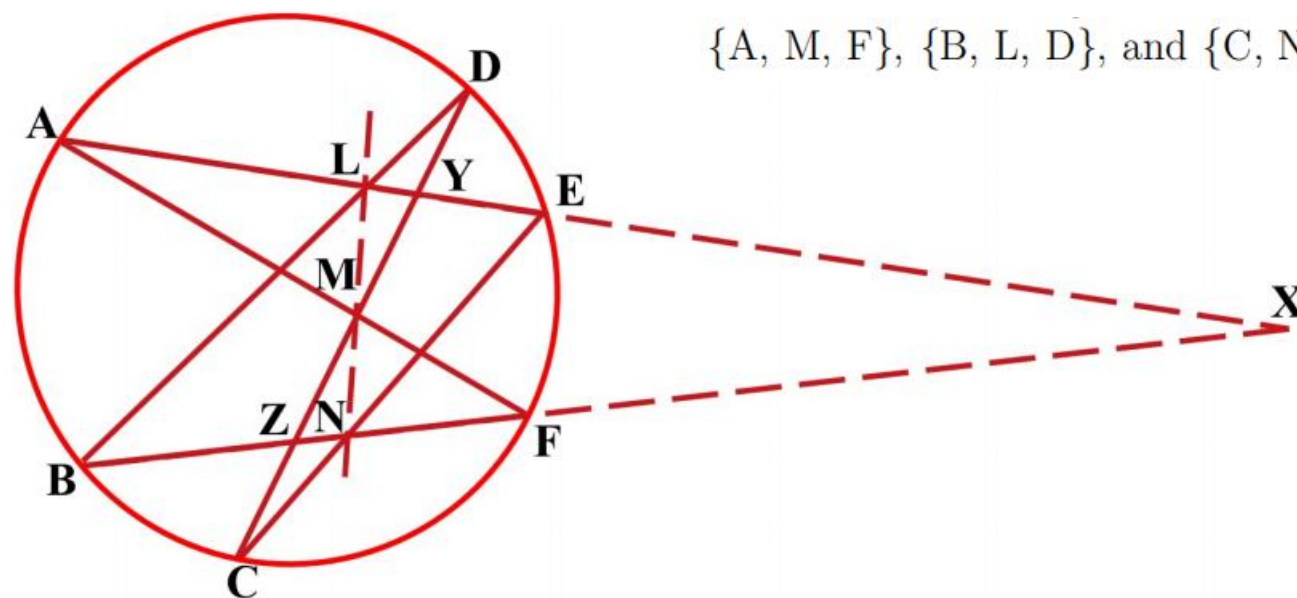


$$|E_1P| = \sqrt{|O_1P|^2 - |O_1E_1|^2} = \sqrt{|O_1P|^2 - |O_1P_1|^2} = |P_1P|$$

$$|E_2P| = \sqrt{|O_2P|^2 - |O_2E_2|^2} = \sqrt{|O_2P|^2 - |O_2P_2|^2} = |P_2P|$$

$$|E_1P| + |E_2P| = |P_1P| + |P_2P| = |P_1P_2| \quad (\text{a constant})$$

Pascal Theorem - PG Perspective



$\{A, M, F\}$, $\{B, L, D\}$, and $\{C, N, E\}$

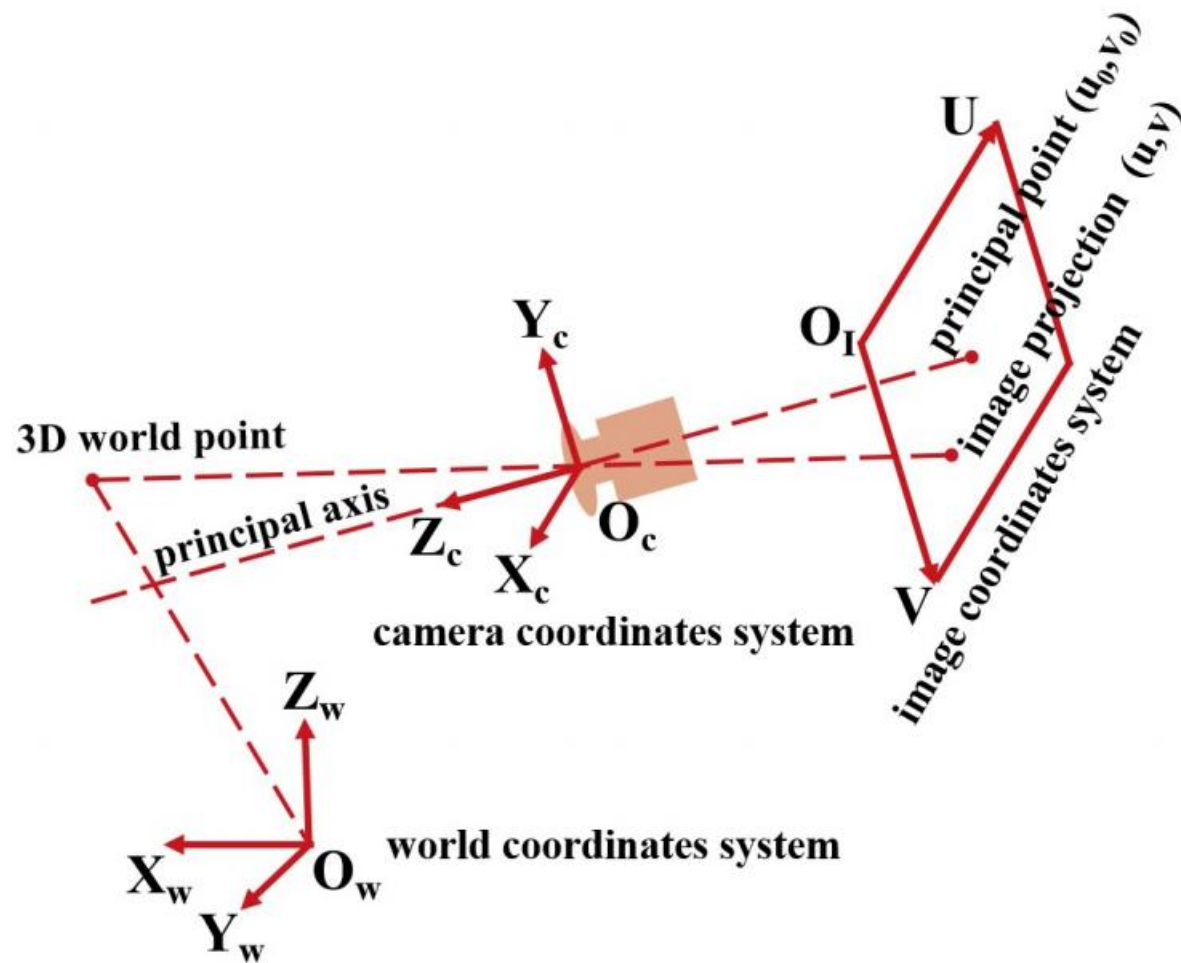
$$\begin{aligned}
 \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} &= 1 & \prod \text{left sides} &= \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} \cdot \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} \\
 \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} &= 1 & &= \frac{|ZM|}{|MY|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XN|}{|NZ|} \cdot \frac{|XF| \cdot |XB|}{|AX| \cdot |EX|} \cdot \frac{|YA| \cdot |YE|}{|CY| \cdot |DY|} \cdot \frac{|ZD| \cdot |ZC|}{|BZ| \cdot |FZ|} \\
 \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} &= 1 & &= \frac{|ZM|}{|MY|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XN|}{|NZ|} = 1,
 \end{aligned}$$



DISCUSSION

Q & A

World, Camera, Image Coordinates Systems



world coordinates system
(WCS)

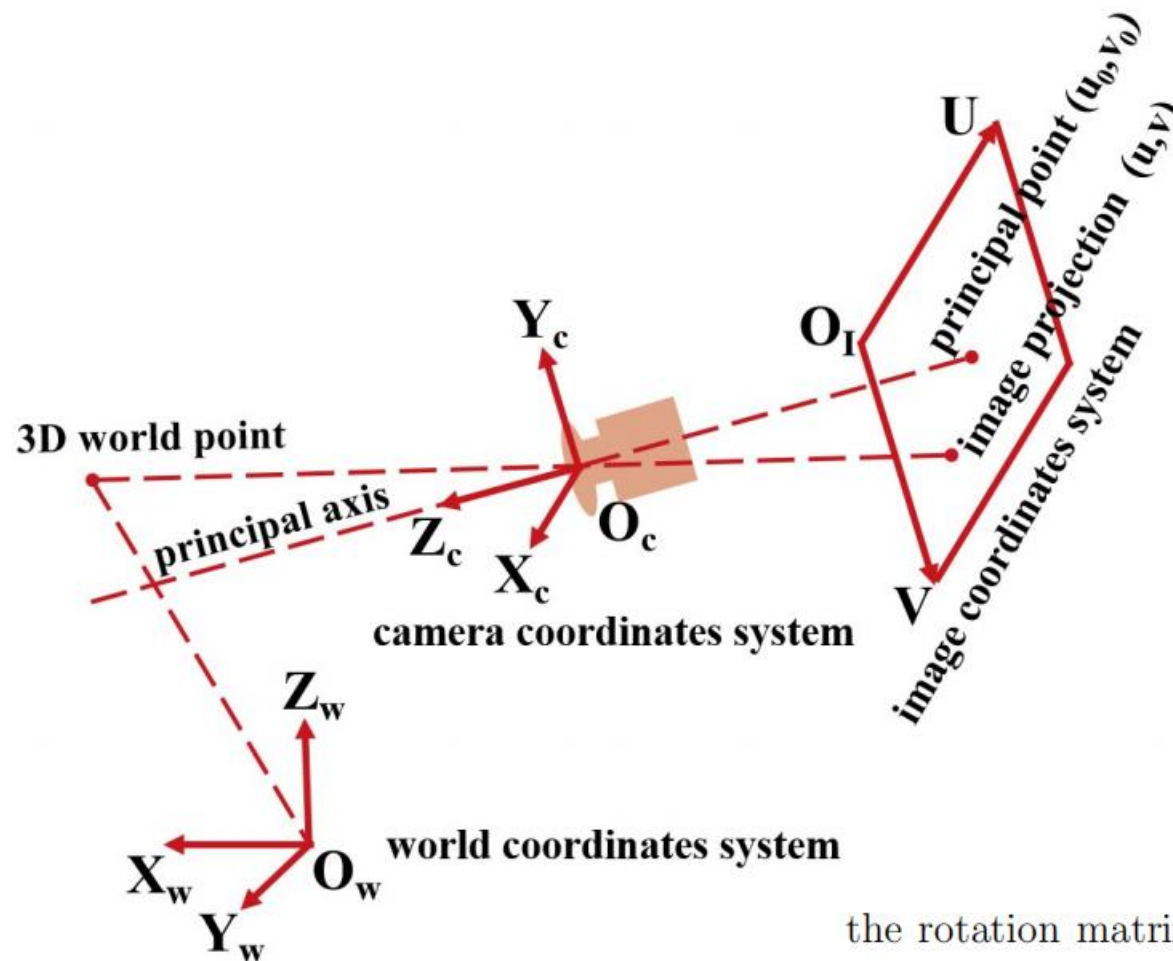


camera coordinates system
(CCS)



image coordinates system
(ICS)

World CS to Camera CS



world coordinates system
(WCS)



camera coordinates system
(CCS)

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = R_{wc} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + T_{wc}$$

$$P_c = R_{wc}P_w + T_{wc}$$

the rotation matrix R_{wc} and the translation vector T_{wc}

camera extrinsic parameters

Basic 3D Rotations



$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$\mathbf{R}_x(\theta) = \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

Properties of the Rotation Matrix

Orthonormality

$$\mathbf{R}^T \mathbf{R} = \mathbf{R} \mathbf{R}^T = \mathbf{I}$$

$$\mathbf{R} \equiv \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\begin{bmatrix} r_{1j} & r_{2j} & r_{3j} \end{bmatrix} \begin{bmatrix} r_{1j} \\ r_{2j} \\ r_{3j} \end{bmatrix} = r_{1j}^2 + r_{2j}^2 + r_{3j}^2 = 1, \quad j \in \{1,2,3\}$$

$$\begin{bmatrix} r_{i1} & r_{i2} & r_{i3} \end{bmatrix} \begin{bmatrix} r_{i1} \\ r_{i2} \\ r_{i3} \end{bmatrix} = r_{i1}^2 + r_{i2}^2 + r_{i3}^2 = 1, \quad i \in \{1,2,3\}$$

$$\begin{bmatrix} r_{1j} & r_{2j} & r_{3j} \end{bmatrix} \begin{bmatrix} r_{1k} \\ r_{2k} \\ r_{3k} \end{bmatrix} = r_{1j}r_{1k} + r_{2j}r_{2k} + r_{3j}r_{3k} = 0, \quad j \neq k$$

$$\begin{bmatrix} r_{i1} & r_{i2} & r_{i3} \end{bmatrix} \begin{bmatrix} r_{k1} \\ r_{k2} \\ r_{k3} \end{bmatrix} = r_{i1}r_{k1} + r_{i2}r_{k2} + r_{i3}r_{k3} = 0, \quad i \neq k$$

$$\mathbf{R}^{-1} = \mathbf{R}^T = \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix}$$

Properties of the Rotation Matrix

Orthonormality

$$\mathbf{R}^T \mathbf{R} = \mathbf{R} \mathbf{R}^T = \mathbf{I}$$

$$\mathbf{R} \equiv \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

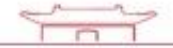
How many degrees of freedom (DoF) are there in the rotation matrix ?

$$r_{1j}^2 + r_{2j}^2 + r_{3j}^2 = 1, \quad j \in \{1, 2, 3\}$$

$$r_{1j}r_{1k} + r_{2j}r_{2k} + r_{3j}r_{3k} = 0, \quad j \neq k$$

$$\text{DoF} = 9 - 6 = 3$$

World CS to Camera CS



$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{R}_{wc} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \mathbf{T}_{wc}$$

If $\mathbf{T}_{wc} = \mathbf{0}$ then $\frac{x_c^2 + y_c^2 + z_c^2}{x_w^2 + y_w^2 + z_w^2} = ?$

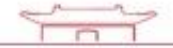
world coordinates system
(WCS)



camera coordinates system
(CCS)

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{0}; \mathbf{R}_{wc} = \begin{bmatrix} \sqrt{0.5} & -\sqrt{0.5} & 0 \\ \sqrt{0.5} & \sqrt{0.5} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \mathbf{T}_{wc} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = ?$$

3D CS-to-CS Transformation Concatenation



$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \mathbf{R}_{12} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \mathbf{T}_{12}$$

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \mathbf{R}_{23} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} + \mathbf{T}_{23}$$

$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{23} & \mathbf{T}_{23} \\ & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{23} & \mathbf{T}_{23} \\ & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$

3D CS-to-CS Transformation Inverse



$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \mathbf{R}_{12} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \mathbf{T}_{12}$$

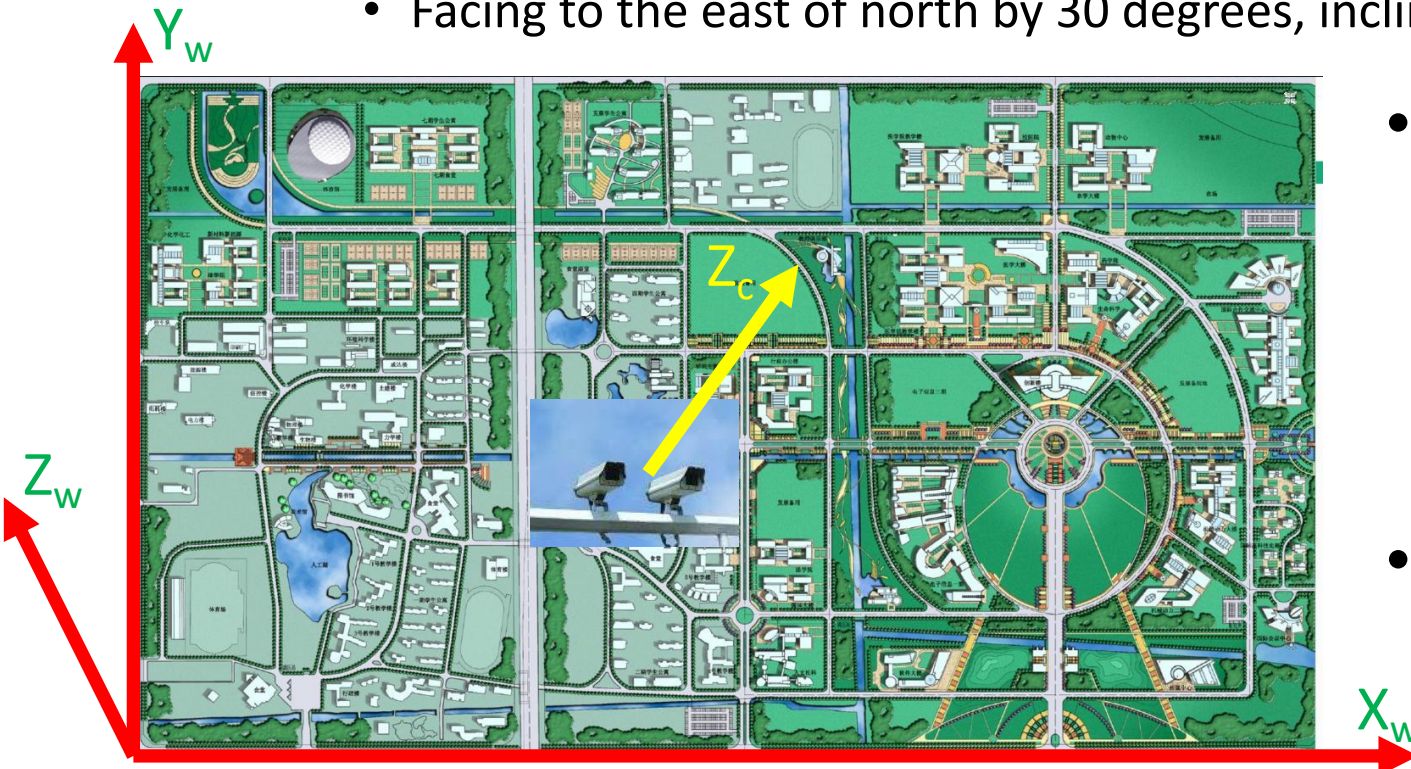
$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12}^T & -\mathbf{R}_{12}^T \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix}$$

Comprehensive Example

- Street-View Camera

- “MapForUI.jpg” : 803x1404 pixels, each pixel representing 1.7 meters in the WCS
- Camera configuration
 - Located at (800, 600) meters, at a height of 10 meters
 - Facing to the east of north by 30 degrees, inclining downward by 15 degrees



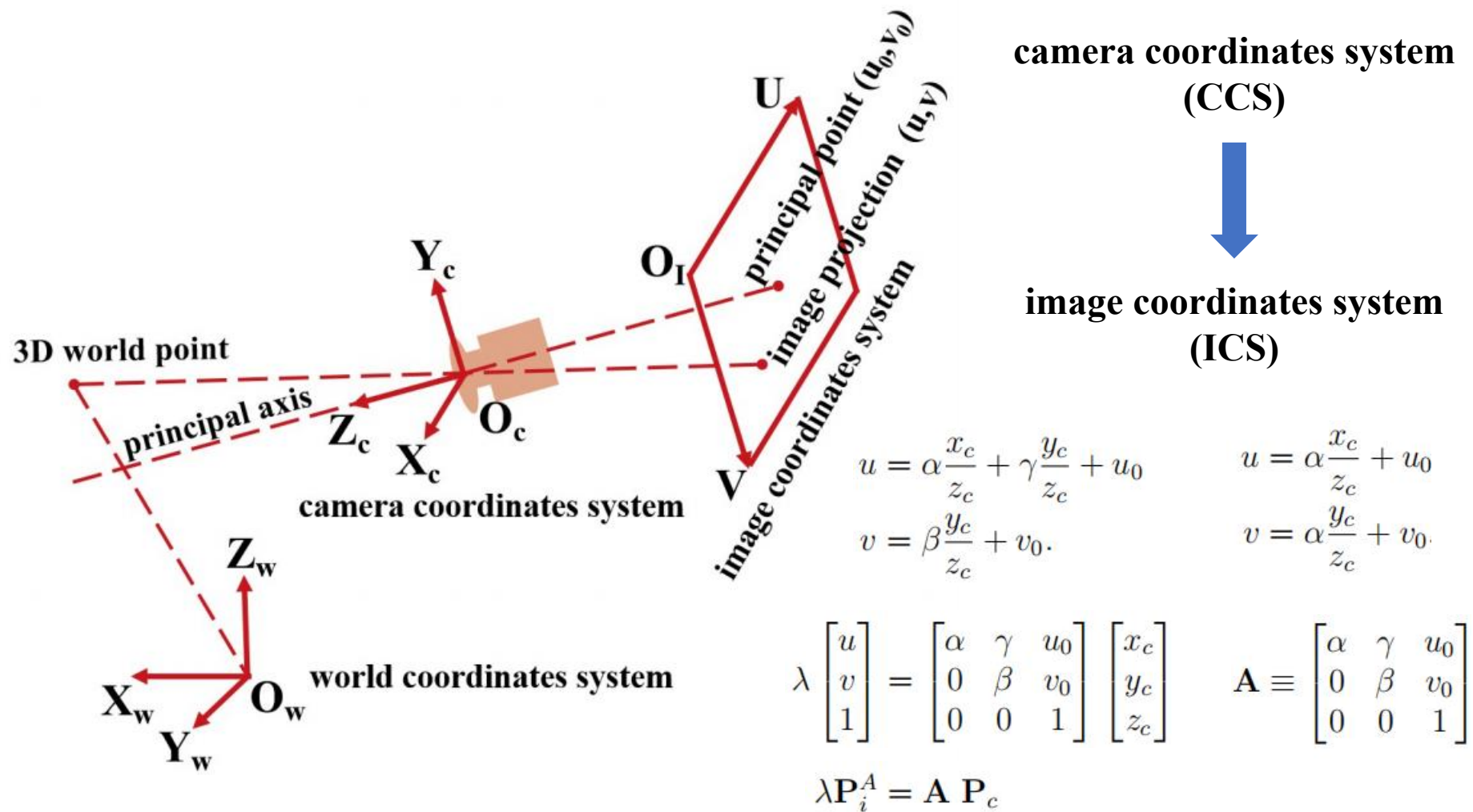
- Intermediate-Pose Approach

- Removal of inclination
- Making the Z-axis upward
- Alignment with the WCS
- Final translation

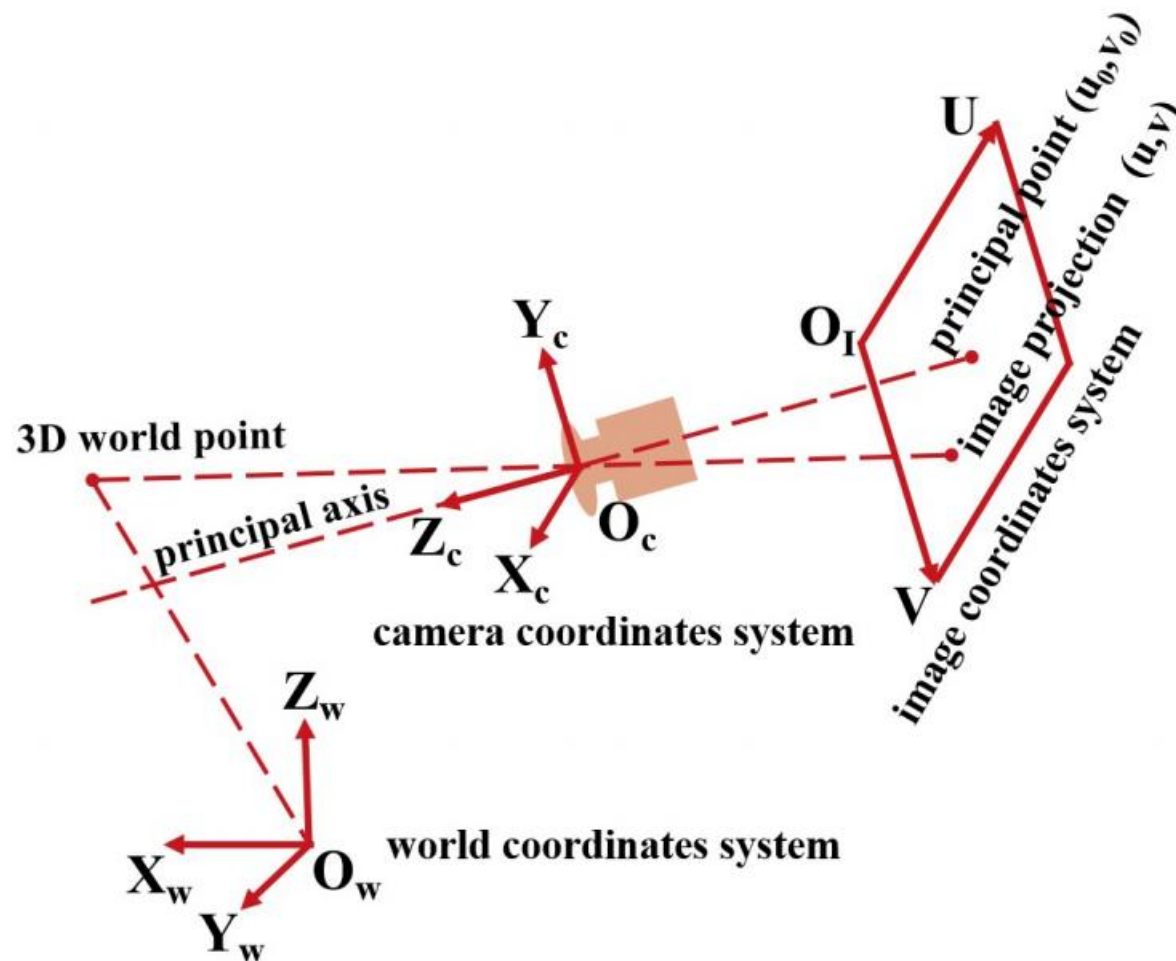
- Base-Vector Approach

- Project each base vector

Camera CS to Image CS



Camera CS to Image CS



camera coordinates system
(CCS)



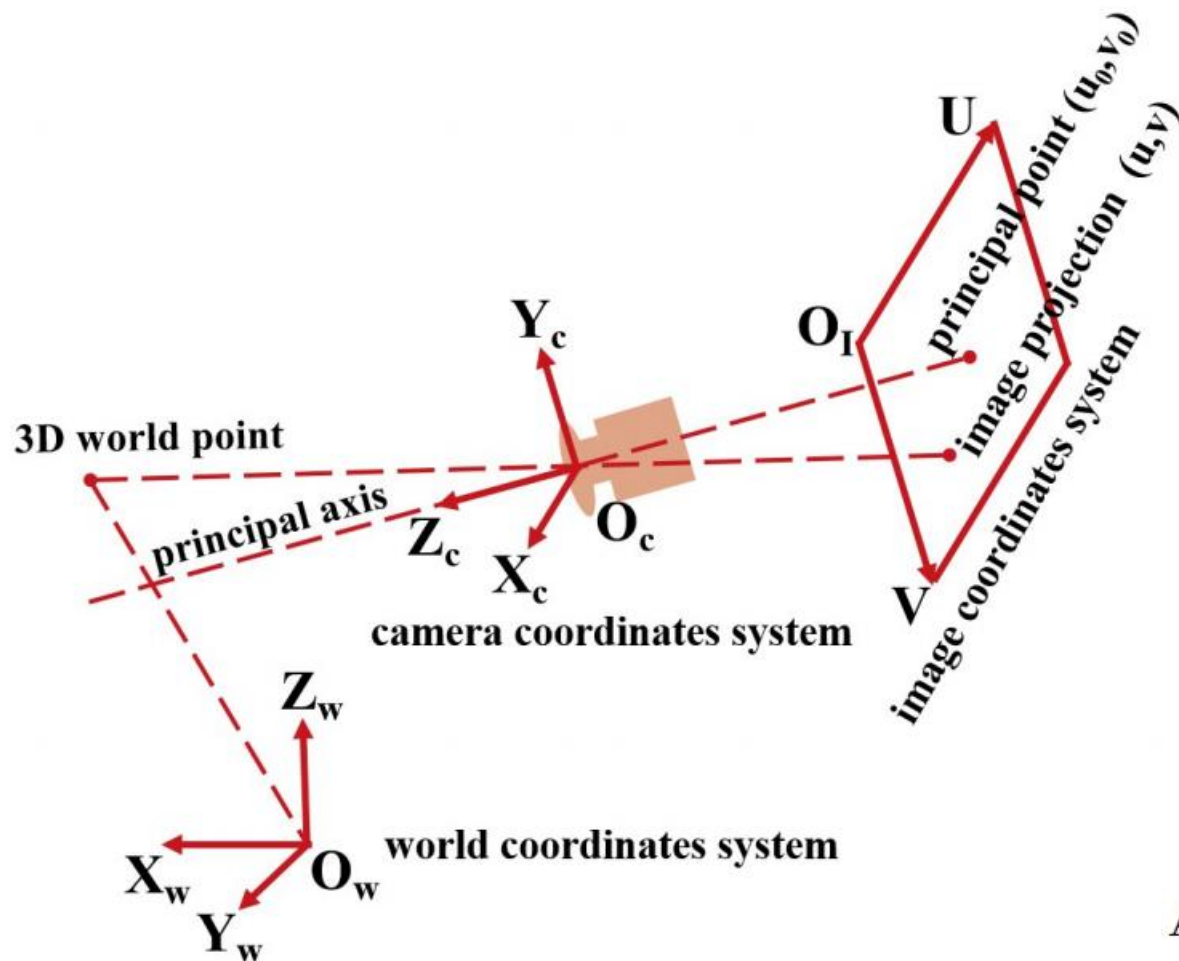
image coordinates system
(ICS)

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} \mathbf{P}_c$$

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \lambda \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Camera CS to Image CS



camera coordinates system
(CCS)



image coordinates system
(ICS)

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} \mathbf{P}_c$$

$\mathbf{A}$

camera intrinsic matrix

$\alpha, \beta, \gamma, u_0,$ and v_0

camera intrinsic parameters

Pay attention to signs in practice

Perspective Mapping

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{R}_{wc} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \mathbf{T}_{wc}$$

$$\mathbf{P}_c = \mathbf{R}_{wc} \mathbf{P}_w + \mathbf{T}_{wc}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} \mathbf{P}_c$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \mathbf{P}_w^A$$

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\mathbf{P}_c = [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \mathbf{P}_w^A$$

**world coordinates system
(WCS)**

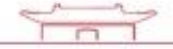


**image coordinates system
(ICS)**

$$\mathbf{M} \equiv \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \equiv \mathbf{A} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}]$$

perspective mapping matrix

Perspective Mapping



$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \mathbf{P}_w^A$$

$$\mathbf{M} \equiv \begin{bmatrix} \mathbf{M}_{1,:} \\ \mathbf{M}_{2,:} \\ \mathbf{M}_{3,:} \end{bmatrix} \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix}$$

perspective mapping matrix

$$u = \frac{\mathbf{M}_{1,:} \mathbf{P}_w^A}{\mathbf{M}_{3,:} \mathbf{P}_w^A} = \frac{m_{11}x_w + m_{12}y_w + m_{13}z_w + m_{14}}{m_{31}x_w + m_{32}y_w + m_{33}z_w + m_{34}}$$
$$v = \frac{\mathbf{M}_{2,:} \mathbf{P}_w^A}{\mathbf{M}_{3,:} \mathbf{P}_w^A} = \frac{m_{21}x_w + m_{22}y_w + m_{23}z_w + m_{24}}{m_{31}x_w + m_{32}y_w + m_{33}z_w + m_{34}}$$

**world coordinates system
(WCS)**



**image coordinates system
(ICS)**

Revisit Common Sense of Projective Geometry



Vanishing point

$$L_w : \mathbf{P}_w(t) \equiv [x_w^0 + x_w t \quad y_w^0 + y_w t \quad z_w^0 + z_w t]^T$$

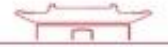
$$u(t) = \frac{\mathbf{M}_{1,:} \mathbf{P}_w^A(t)}{\mathbf{M}_{3,:} \mathbf{P}_w^A(t)} = \frac{(m_{11}x_w^0 + m_{12}y_w^0 + m_{13}z_w^0 + m_{14}) + (m_{11}x_w + m_{12}y_w + m_{13}z_w)t}{(m_{31}x_w^0 + m_{32}y_w^0 + m_{33}z_w^0 + m_{34}) + (m_{31}x_w + m_{32}y_w + m_{33}z_w)t}$$

$$v(t) = \frac{\mathbf{M}_{2,:} \mathbf{P}_w^A(t)}{\mathbf{M}_{3,:} \mathbf{P}_w^A(t)} = \frac{(m_{21}x_w^0 + m_{22}y_w^0 + m_{23}z_w^0 + m_{24}) + (m_{21}x_w + m_{22}y_w + m_{23}z_w)t}{(m_{31}x_w^0 + m_{32}y_w^0 + m_{33}z_w^0 + m_{34}) + (m_{31}x_w + m_{32}y_w + m_{33}z_w)t}$$

As $t \rightarrow \infty$, the $[u(t) \quad v(t)]^T$ of all the parallel lines approach the common limit

$$\begin{cases} u(t) & \rightarrow \frac{m_{11}x_w + m_{12}y_w + m_{13}z_w}{m_{31}x_w + m_{32}y_w + m_{33}z_w} \\ v(t) & \rightarrow \frac{m_{21}x_w + m_{22}y_w + m_{23}z_w}{m_{31}x_w + m_{32}y_w + m_{33}z_w} \end{cases}$$

Revisit Common Sense of Projective Geometry



- *Invariance 1*: A point after perspective mapping (or perspective transform) is still a point.
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DISCUSSION

Q & A